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Patterns of trust and subjective well-being across societies: the role of long-term versus short-term orientation

Jiansong Zheng[1] and Tao Zhang[1*]

Abstract

Existing literature remains limited regarding the various associations between social trust and subjective well-being across different time-based cultural norms. To address this gap, this study examines the moderating role of societal long-term versus short-term orientation in the individual -level relationships between individuals' subjective well-being and interpersonal trust (i.e. particularized and generalized trust). Multilevel models are conducted using a sample of 120,682 individuals across 80 societies, based on the joint European Values Study and World Values Survey dataset. The results indicate that both types of interpersonal trust show positive correlations with subjective well-being across societies. In societies with more long-term orientation, both forms of interpersonal trust exhibit a stronger and positive correlation with subjective well-being. In contrast, in short-term orientation societies, both forms of interpersonal trust show a weaker yet still positive correlation with well-being. These findings underscore the importance of societal time orientations in shaping the relationship between interpersonal trust and well-being.

Keywords Particularized trust, Generalized trust, Subjective well-being, Long-term versus short-term orientation

Introduction

Subjective well-being, pertains to individuals' perceptions of their overall quality of life and emotional experiences [1, 2]. Well-being is essential not only for personal fulfillment but also for the advancement of societal progress and flourishing [3]. Research indicates that individuals' subjective well-being is correlated with numerous factors, such as socioeconomic status, social relationships, and cultural values [2, 4, 5]. As social relationships are widely recognized as strong predictors of well-being, social trust, which is closely tied to interpersonal relationships, has also been demonstrated to be a significant contributor to subjective well-being [6]. In a global context, Q Guo, et al. [7] found social trust more strongly associated with well-being in individualistic cultures. Building on this, this study extends the work of Q Guo et al. [7] by incorporating an additional Hofstede cultural dimension: long-term versus short-term orientation.

Literature review and hypotheses

Interpersonal trust and subjective well-being

Interpersonal trust reflects individuals' confidence in the reliability and benevolence of others, beyond personal relationships to encompass broader societal networks [8]. In this case, interpersonal trust manifests in various forms, including particularized and generalized trust, which are categorized based on the varying social distance of the trust groups [9]. Particularized trust refers

*Correspondence:
Tao Zhang
taozhang@mpu.edu.mo
[1]Faculty of Humanities and Social Sciences, Macao Polytechnic University, Macau, China

to trust within close-knit social circles, such as family, neighbors, and friends [10]. Generalized trust encompasses trust in unfamiliar others with "trustee" as strangers, foreigners, and people with different religions [11]. Social relationships, closely linked to overall well-being, become increasingly distant from particularized trust to generalized trust and outgroup trust [9]. Collectively, these forms of interpersonal trust foster an environment conducive to cooperation, reciprocity, and collective action, thereby enhancing societal cohesion and well-being [7].

A large body of literature has examined the associations between trust and subjective well-being [4, 5, 7, 12–15]. Theories such as social capital theory [16, 17] and social exchange theory [18] offer conceptual frameworks elucidating the mechanisms through which interpersonal trust contributes to subjective well-being. According to these theories, interpersonal trust fosters cooperative behaviors, reciprocity, and the exchange of social resources, thereby enhancing individuals' psychological functioning and overall well-being [14]. Overall, the empirical evidence underscores interpersonal trust as a robust and positive predictor of subjective well-being in Serbia [19]. Extending this associations to more countries, C Glatz and A Eder [13] found a weak and positive correlation between social trust and subjective well-being over time using longitudinal data across 36 European countries. Similarly, Q Guo, et al. [7] found positive associations between all types of interpersonal trust and subjective well-being, and they also found trust more strongly associated with well-being in individualistic cultures. These findings highlight the significance of interpersonal trust as a contributor of well-being, suggesting that higher levels of interpersonal trust contribute to greater subjective well-being. Based on this, we developed a series of hypotheses:

H1: Individuals' interpersonal trust (i.e. particularized and generalized trust) positively predicts subjective well-being across societies.

The moderating role of long-term versus short-term orientation

Cultural factors may play a critical role in shaping associations between interpersonal trust and subjective well-being. For instance, Q Guo et al. [7] found that interpersonal trust is more strongly associated with well-being in individualistic cultures. Cross-cultural research suggests a potential overlap between long-term versus short-term orientation and individualism/collectivism [20]. However, some studies distinguish their differences by focusing on several aspects including time perspective and social relationship [10, 21]. For instance, long-term orientation places more emphasis on future outcomes, whereas individualism is not necessarily tied to a specific time frame [22]. Since time preferences are correlated with trust and well-being [10, 23, 24], it is important to explore the moderating role of long-term versus short-term orientation in the relationships between interpersonal trust and subjective well-being.

Societal long-term versus short-term orientation may moderate the relationship between interpersonal trust and subjective well-being. In short-term-oriented societies like Serbia, where people focus more on the present moment and embrace change, a positive association between interpersonal trust and subjective well-being has been observed [10, 19]. A European countries, most of which score moderately on the long-term versus short-term orientation dimension, also exhibit a positive correlation between social trust and well-being [13]. In contrast, there are mixed results in societies with long-term orientation where people place value on perseverance, thriftiness, and preparation for future challenges [25]. For instance, South Korea is a long-term orientation society, and people there experience a positive association between happiness and particularized trust, but an insignificant individual-level relationship between generalized trust and well-being [26]. However, C Bai, et al. [15] used Chinese sample to poit a positive correlation between well-being and trust in strangers, but an insignificant relationship between subjective well-being and particularized trust (e.g., trust in family and acquaintances) in China where people follow more on long-term orientation. These mixed findings suggest that societal long-term versus short-term orientation may influence associations between well-being and varying patterns of interpersonal trust. Consequently, this study aims to explore the moderating role of long-term versus short-term orientation in shaping the associations between different patterns of interpersonal trust and subjective well-being. In this case, the following hypotheses were formulated:

H2: Societal long-term versus short-term orientation moderates the associations between individuals' subjective well-being and interpersonal trust.

H2a: Societal long-term versus short-term orientation moderates the associations between individuals' subjective well-being and particularized trust.

H2b: Societal long-term versus short-term orientation moderates the associations between individuals' subjective well-being and generalized trust.

This study

This study builds upon the work of Q Guo et al. [7], with a key distinction being the inclusion of the moderating role of long-term versus short-term orientation. While societal norms regarding long-term versus short-term orientation influence individuals' time preferences and are closely tied to trust and well-being, this cultural dimension has received relatively little attention in previous cross-culture research. By extending Q Guo et al.'s [7] study, this research incorporates more countries and regions to explore the associations between different patterns of interpersonal trust and subjective well-being. Given the exploratory nature of Hypothesis 2, the directions of the moderating effects of long-term versus short-term orientation are not predetermined. The research framework is illustrated in Fig. 1.

Q Guo, et al. [7] emphasized the significant role of individualism/collectivism in subjective well-being relevant to trust, and we have included it as a control variable in our analysis. Additionally, considering that individuals in societies with a higher level on the Human Development tend to show higher degrees of well-being, we included this factor in our models as a control variable [27]. We also controlled for demographic factors affecting subjective well-being, including relative income, gender, age, and education level [28, 29].

Data and method

Data and softwares

We used the joint EVS-WVS 2017–2022 dataset Version 5.0.0 to construct the individual-level data. The joint EVS-WVS dataset consists of EVS5 (European Values Study Wave 5) and WVS7 (World Values Survey Wave 7) [30]. A total of 153,716 respondents from 92 countries and regions participated in the joint EVS-WVS. A total of 120,682 respondents from 80 countries and regions participated in this study by matching the joint WVS-EVS dataset and country attribute data from Hofstede Insights and the Human Development Report. R version 4.3.2 was used for data cleaning, descriptive analysis, and data visualization. Stata Version 15 was used to obtain the results of multilevel models. This study was preregistered on the Open Science Framework (OSF), and the associated data and codes have been uploaded [31].

Variables

Subjective well-being was measured using both variables including happiness, and life satisfaction extracted from the joint EVS-WVS dataset items "Taking all things together, would you say you are very happy, quite happy, not very happy, or not at all happy?", "All things considered, how satisfied are you with your life as a whole these days?". Happiness and life satisfaction are rated on a 4-point Likert scale and 10-point Likert scale, respectively. The responses for happiness were reverse-scored, while life satisfaction responses remained consistent with the original values. The mean score was utilized to gauge respondents' well-being, with higher values indicating greater subjective well-being.

Particularized trust (PT) was assessed by inquiring about respondents' levels of trust in three target groups including their family, neighborhood, and people they know personally [32]. The responses were reverse scored and the final scores were rated on a 4-point Likert scale, ranging from 1 (Do not trust at all) to 4 (Trust completely). Average scores were computed to gauge respondents' particularized trust, with higher values indicating greater levels of trust.

Generalized trust (GT) was assessed by measuring participants' trust in people they meet for the first time, people of another religion, and people of another nationality

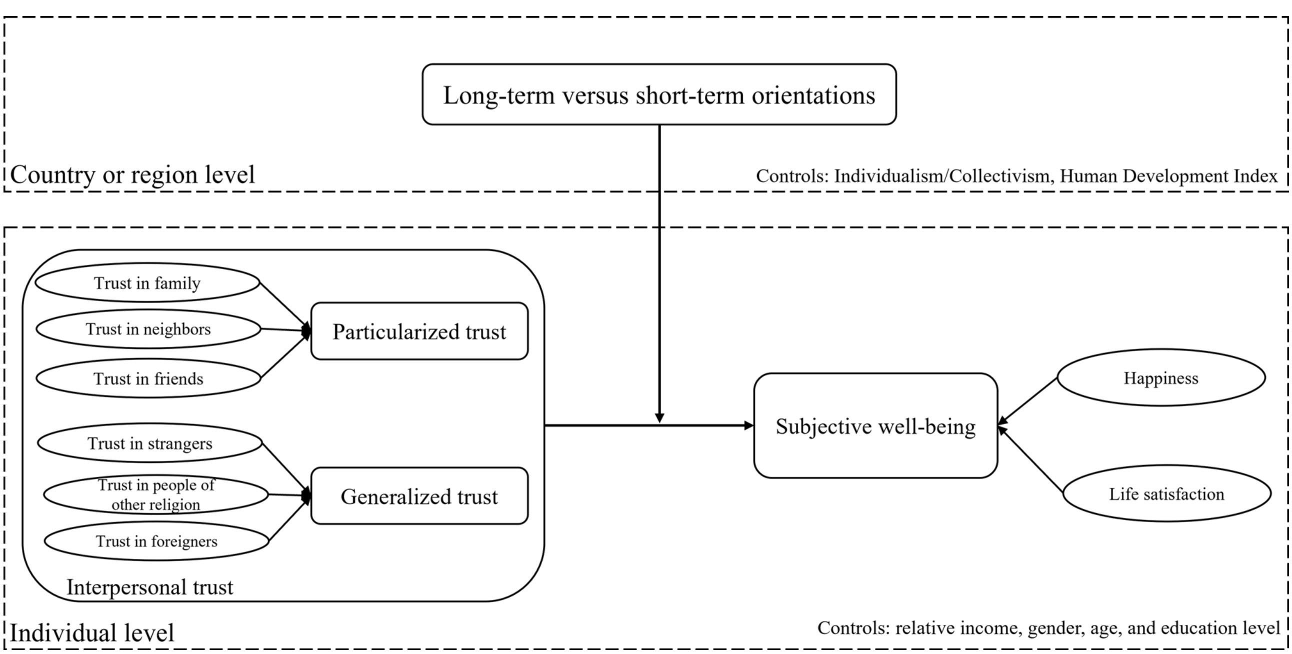

Fig. 1 Research framework

[33]. The responses were reverse coded. The final score was rated on a 4-point Likert scale, ranging from 1 (Do not trust at all) to 4 (Trust completely). The higher values indicate higher levels of their generalized trust.

The joint EVS-WVS dataset also contains individuals' demographic information including *relative income* using a 10-point Likert scale with higher values representing respondents' self-reported higher income relative to other people in their country or region, *gender* coded as "1: male, 0: female", *age* as a continuous variable, and *education level* coded as "0: less than primary, 1: primary, 2: lower secondary, 3: upper secondary, 4: post-secondary or non-tertiary, 5: short-cycle tertiary, 6: bachelor or equivalent, 7: master or equivalent, 8: doctoral or equivalent".

Long-term versus short-term orientation (LTO/STO) is a country or region-level variable collected from the official Hofstede Cultural Dimensions website [34, 35]. LTO/STO is measured on a scale ranging from 0 to 100. The lower scores indicate people in the country or region tend towards more short-term orientation, while higher scores represent that residents in the country or region tends towards more long-term orientation.

Individualism/Collectivism (IND/COL) is a country- or region-level variable derived from Hofstede's Cultural Dimensions[1] [34, 35]. The IND/COL is a scale ranging from 0 to 100, where higher scores indicate people's greater tendency towards individualism, while lower scores reflect residents' stronger inclination towards collectivism within a given country or region.

The Human Development Index (HDI) derived from the Human Development Report 2021/2022 [36], and the HDI score of 2021 were used. HDI is a composite indicator summarizing the average achievement of a country or region across three dimensions: life expectancy, education, and per capita income. HDI is ranged from 0 to 1 with higher scores representing higher levels of human development of a country or region. The missing values of HDI from three countries and regions including Taiwan, Turkey, and Puerto Rico, were imputed using data from their official statistical departments.

Analysis strategy

A null model, excluding all the independent variables, was conducted to calculate the intraclass correlation coefficient (ICC). ICC is an indicator used to assess the need for a multilevel model. If the ICC exceeds 0.059, a multilevel model is necessary to account for international differences in individual-level subjective well-being [37].

Both main multilevel models with random slopes and intercepts were conducted to examine the association between subjective well-being and interpersonal trust including particularized and generalized trust. Main models were added with individual-level control variables including relative income, gender, age, and education level. All continuous variables were standardized prior to analysis.

Two full multilevel models with random slopes and intercepts were conducted by adding country or region-level variables including long-term versus short-term orientation, individualism-collectivism, and the Human Development Index. Also, another two full multilevel models were utilized to observe the coefficients of cross-level interaction between long-term versus short-term orientation and interpersonal trust (i.e. particularized and generalized trust). The individual-level independent and control variables in the full models remain consistent with those in the main models. Simple slope tests for multilevel models with the cross-level interaction item were conducted to further investigate the significant moderating role of long-term versus short-term orientation in the association between interpersonal trust and subjective well-being [38].

Results

Figure 2 descriptively plots the associations between subjective well-being, interpersonal trust and long-term versus short-term orientation. In countries and regions with long-term orientation, such as the Nordic countries, individuals consistently report higher levels of subjective well-being and interpersonal trust. Conversely, in countries and regions characterized by short-term orientation, the relationship between subjective well-being and interpersonal trust appears more heterogeneous, with varying correlations across different types of trust. To better understand these dynamics, it is essential to conduct statistical modeling to clarify the underlying mechanisms.

We calculated the estimations of null models with subjective well-being as the dependent variable. The ICC value of the null model is 0.110 (95%CI = [0.084, 0.146]), greater than 0.059, indicating it is necessary to conduct multilevel models to analyze the national differences in residents' subjective well-being.

Table 1 shows the results of main models. The coefficients of interpersonal trust (i.e. particularized and generalized trust) are all significant, indicating they are significant predictors of subjective well-being, in favor of H1. Individual-level control variables can significantly predict subjective well-being. For instance, higher relative income and education levels are positively associated with greater subjective well-being. Females tend to report lower levels of subjective well-being compared to males. Age may negatively predict subjective well-being.

Table 2 shows the estimations of full models. The country- or region-level coefficients of long-term versus short-term orientation remain significant at the 5% level in models where particularized trust is the dependent variable, but these coefficients are insignificant in

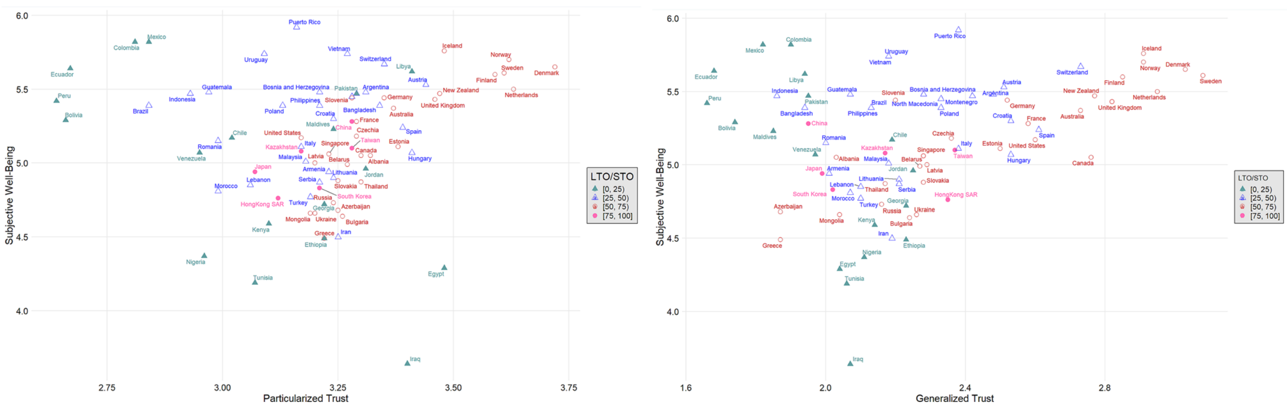

Fig. 2 Scatterplot for the association between subjective well-being and particularized trust (left) as well as generalized trust (right), categorized by LTO/STO. Note. LTO/STO represent long-term versus short-term orientation

Table 1 Estimations of main multilevel models with subjective well-being as the dependent variable

		Model 1	Model 2
Individual-level coefficient	Particularized trust	0.192***	
		(0.010)	
	Generalized trust		0.082***
			(0.009)
	Relative income	0.196***	0.204***
		(0.014)	(0.015)
	Gender	-0.052***	-0.046***
		(0.008)	(0.008)
	Age	-0.035**	-0.022
		(0.012)	(0.013)
	Education level	0.022**	0.019*
		(0.007)	(0.008)
Akaike's information criterion		317245.4	320798.9

Notes. 1. Standard errors in parentheses

2. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$

3. $N = 120{,}682$

4. The number of countries and regions is 80

models with generalized trust as the dependent variable. Although the Human Development Index is positively correlated with individuals' subjective well-being, it does not significantly predict it. The addition of country- or region-level variables provided only a marginal improvement in model explanatory power, as indicated by minimal changes in values of Akaike Information Criterion (AIC; $\Delta AIC_{model\ 3-1}$ = -0.6; $\Delta AIC_{model\ 4-2}$ = -0.6) [39].

Table 2 also presents the results of full models with the cross-level interaction terms. The interaction terms between different types of interpersonal trust and long-term versus short-term orientation are significantly positive, indicating that long-term versus short-term orientation siginificantly moderates the association between interpersonal trust and subjective well-being, in favor of H2. The inclusion of cross-level interaction terms improved the explanatory power of the models, as reflected by lower AIC values ($\Delta AIC_{model\ 5-3}$ = -18.4; $\Delta AIC_{model\ 6-4}$ = -21.3), indicating a better model fit with the addition of these interactions.

The simple slope test was conducted in order to further investigate the cross-level moderation role. Figure 3 shows the results of the simple slope test. Long-term versus short-term orientation scores were divided into quartiles to represent short-term and long-term orientation. Interpersonal trust and subjective well-being are taken as the mean plus and minus three standard deviations, and the global sample can be covered.

The simple slope test results show that in countries and regions with long-term oritentation, the coefficients of people's particularized trust on their subjective well-being were higher, suggesting that individuals with higher levels of particularized trust experience greater subjective well-being in the countries and regions where people tend towards more long-term orientation. H2a was supported. For results with generalized trust as the independent variable, the moderating pattern remain consistent with models with particularized trust as the independent variable, in favor of H2b.

Discussion

This study identified positive associations between subjective well-being and various patterns of interpersonal trust. In other words, both particularized and generalized trust exhibit positive correlations with subjective well-being across societies. The coefficients of particularized trust were larger than those of generalized trust, potentially supporting the notion that particularized trust is a stronger predictor of well-being [40]. Nevertheless, both the coefficients remain consistent and positive in direction, and this finding aligns with a recent meta-analysis which concluded that trust types do not moderate the association between trust and well-being [41]. Furthermore, these positive associations across societies are aligned with several previous studies using multinational samples [7, 13]. In this case, these findings reinforce the

Table 2 Estimations of full multilevel models with subjective well-being as the dependent variable

		Model 3	**Model 4**	**Model 5**	**Model 6**
Individual-level coefficient	Particularized trust	0.192***		0.198***	
		(0.010)		(0.009)	
	Generalized trust		0.082***		0.087***
			(0.009)		(0.008)
	Relative income	0.196***	0.204***	0.196***	0.204***
		(0.014)	(0.015)	(0.014)	(0.015)
	Gender	-0.052***	-0.046***	-0.052***	-0.046***
		(0.008)	(0.008)	(0.008)	(0.008)
	Age	-0.035**	-0.022	-0.035**	-0.022
		(0.012)	(0.013)	(0.012)	(0.013)
	Education level	0.021**	0.019*	0.021**	0.019*
		(0.007)	(0.008)	(0.007)	(0.008)
Country and region-level coefficient	LTO/STO	-0.082*	-0.069	-0.081*	-0.067
		(0.040)	(0.040)	(0.040)	(0.040)
	IND/COL	0.019	0.029	0.018	0.029
		(0.064)	(0.062)	(0.064)	(0.062)
	HDI	0.092	0.084	0.092	0.083
		(0.065)	(0.064)	(0.065)	(0.064)
Cross-level ineraction item coefficient	Particularized trust × LTO/STO			0.038***	
				(0.008)	
	Generalized trust × LTO/STO				0.038***
					(0.007)
Akaike's information criterion		317244.8	320798.3	317226.4	320777.0

Notes. (1) Standard errors in parentheses. (2) LTO/STO means long-term versus short-term orientation culture, IND/COL is individualism/collectivism, and HDI indicates Human Development Index. (3) $^{***}p<0.001$, $^{**}p<0.01$, $^{*}p<0.05$. (4) $N=120{,}682$. (5) The number of countries and regions is 80

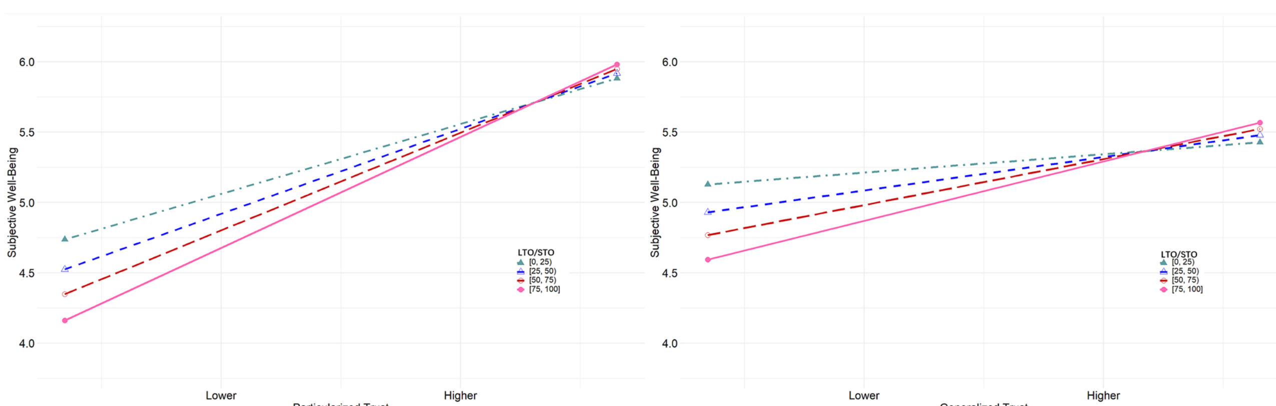

Fig. 3 Results of simple slope test with generalized trust (left) and generalized trust (right) predicting subjective well-being, moderated by long-term versus short-term orientation. Note. LTO/STO represent long-term versus short-term orientation

academic consensus regarding the universal positive relationship between trust and well-being around the world [7, 42, 43].

The significant and positive cross-level interaction item coefficient between interpersonal trust and long-term versus short-term orientation provides valuable insights into the role of cultural values in shaping the relationships between interpersonal trust and subjective well-being. In societies with long-term orientation, characterized by a focus on stability, tradition, and intergenerational cooperation, the individual-level association between interpersonal trust and subjective well-being may be more positive [25]. Individuals in these societies may place greater emphasis on enduring social bonds and collective goals, leading to a more gradual and enduring effect of interpersonal trust on their subjective well-being [44]. In societies with more short-term orientation, marked by adaptability, flexibility, and immediate gratification, the association between interpersonal trust and subjective well-being may be positive but weaker [45]. In such contexts, individuals may place significant reliance on immediate social interactions and trust, leading to a direct and immediate association between interpersonal trust and subjective well-being [46]. However, this relationship may be subject to change over time [47]. These interpretations underscore the importance of accounting for time-related cultural contexts when examining the relationship between interpersonal trust and subjective well-being, as societal values influence the strength of this relationship.

The coefficient values of two cross-level interaction terms are very similar, indicating consistent moderating effects of long-term and short-term orientation on the relationships between both forms of trust and well-being. For instance, as particularized trust demonstrates a stronger predictor of well-being, this association is more pronounced in societies with long-term orientation. The possible reason is that in societies with more long-term orientation, individuals prioritize enduring, intimate, and close-knit relationships, which align closely with particularized trust and are strongly associated with well-being [6]. Conversely, as generalized trust is a weaker but positive predictor of well-being, this prediction is less salient in societies with short-term orientation. The likely explanation is that in societies with short-term orientation, individuals are less inclined to rely on broad trust to secure their future well-being [10]. These findings potentially underscore the moderating roles of long-term and short-term orientation and types of trust, highlighting how time-based values shape the association between trust and well-being across various trust targets.

This study has several theoretical implications, contributing to both social capital theory and cultural dimension theory. The observed positive association between trust and well-being highlights the adaptability of social capital theory in addressing social issues across different societies [16]. Additionally, the significant moderating role of long-term versus short-term orientation represents an application of cultural dimension theory, particularly to a dimension that has received relatively less attention in prior trust research [34]. The varying strengths of the correlation between different types of trust and well-being in societies with differing levels of long-term versus short-term orientation provide new insights into the role of trust radius in shaping the relationship between trust and well-being under distinct time-based social norms [48].

Practically, these findings have important implications for policymakers, governments, and interventions aimed at promoting well-being within diverse time-based cultural contexts. For instance, in long-term oriented societies, policymakers can prioritize maintaining residents' trust, particularly particularized trust, as it is closely linked to subjective well-being. Given the greater difficulty in restoring trust within such social contexts, policies should focus on preventing fraudulent activities to avoid further erosion of trust. In contrast, in societies with short-term orientation, governments may place relatively less emphasis on maintaining residents' trust, especially generalized trust, since its association with subjective well-being is weaker. Instead, policies can focus on addressing residents' short-term goals and directly enhancing well-being through targeted social welfare programs, thereby meeting immediate needs and fostering social harmony. Overall, our findings offer valuable insights for practical application in the pursuit of enhancing well-being across different time-based cultural contexts.

There are several limitations. First, double-item measures of subjective well-being, may reduce the reliability of this study, and future research can consider more dimensions such as positive affect and negative emotional experience. Second, cross-sectional data used in this study fails to infer causality, and the upcoming investigation can use longitudinal dataset to explore causal associations between trust and well-being. Final, when employing a multicultural sample, it is essential to ensure measurement invariance for constructs like generalized trust [49, 50]. Although the multilevel models with random intercepts employed in this study can partially account for national differences, alternative approaches, such as multi-group confirmatory factor analysis, may offer a more precise solution for addressing measurement invariance.

Conclusion

This study conducted multilevel models by using the joint EVS-WVS dataset (2017–2022) as well as country attribute data. The results show that individuals' interpersonal trust including particularized and generalized trust positively and siginifiantly predicted their subjective well-being. Associations between both types of interpersonal trust and subjective well-being were stronger in societies with long-term orientation compared to those with short-term orientation.

Abbreviations

LTO/STO	Long-Term Versus Short-Term Orientation
IND/COL	Individualism / Collectivism
HDI	Human Development Index
ICC	Intraclass Correlation Coefficient
CI	Cofidence Interval
EVS	European Values Study
WVS	World Values Survey Wave
OSF	Open Science Framework

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Author contributions

JSZ and TZ conceptualized the study. JSZ led the writing of the article with critical comments provided by TZ. TZ revised the manuscript. JSZ oversaw data collection and data cleaning. JSZ conducted the analysis with guidance from TZ. All authors have read and approved the final manuscript.

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Data availability
The data used in this study are available in the Open Science Framework repository at https://osf.io/r235d/.

Declarations

Ethics approval and consent to participate
Not applicable, as this study exclusively utilized secondary data sources.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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